

Algebraic Geometry and Schubert Calculus

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Motivating Problem

How many lines intersect 4 lines in "general position"?

Projective space

Definition

Projective space $\mathbb{P}^n$ is the set of equivalence classes of $\mathbb{C}^{n+1} \setminus \{0\}$ under the equivalence relation $x \sim y \iff x = \lambda y, \lambda \in \mathbb{C} \setminus \{0\}$.

“The set of lines in $\mathbb{C}^{n+1}$ ”

Intuition

Projective space is affine space with an added point at infinity, where parallel lines meet at infinity.

Zariski topology

Topology we use in algebraic geometry.

On $\mathbb{C}^n$

Closed sets are the vanishing points of polynomial ideals in $\mathbb{C}[X_1, X_2, \dots, X_n]$.

On $\mathbb{P}^n$

Closed sets are the vanishing points of *homogeneous* polynomials (e.g. $X_1 + X_2$, $X_1^2 - X_3X_2 + X_2X_2$, $X_1X_5 + X_3X_4 - X_2X_4$).

Grassmannian and Plucker embedding

Definition

The Grassmannian $\text{Gr}(k, n)$ is the set of k -dimensional subspaces of $\mathbb{C}^n$.

Embedding

The Plucker embedding send the Grassmannian into projective space.

Map

$$\text{Gr}(k, n) \rightarrow \mathbb{P}^{\binom{n}{k}-1}, \langle v_1, v_2, \dots, v_k \rangle \mapsto v_1 \wedge v_2 \wedge \dots \wedge v_k$$

Plucker embedding

Linear algebra perspective

Sends a $k \times n$ matrix to the determinants of its $k \times k$ minors.

Example

$$\begin{pmatrix} a & b \\ c & d \\ e & f \\ g & h \end{pmatrix} \mapsto [ad - bc, af - be, ah - bg, cf - de, ch - dg, eh - fg].$$

$\text{Gr}(2, 4)$ closed with ideal $(X_1X_6 - X_2X_5 + X_3X_4)$.

Flags

Definition

A flag

$$V_1 \subset V_2 \subset V_3 \subset \dots \subset V_{n-1} \subset \mathbb{C}^n, \dim(V_i) = i$$

is a sequence of subspaces of increasing dimension, each containing the previous one.

Association to a basis

We can associate a basis $(v_1, v_2, \dots, v_n)$ with a flag where the space V_i is the space generated by the first i basis vectors
 $V_i = \langle v_1, v_2, \dots, v_i \rangle$.

Plucker embedding on flags

We send each space V_i of the flag into the Grassmannian $\text{Gr}(k, n)$.
Then compose with the original Plucker embedding.

$$\begin{array}{ccccccc}
 G/B & \hookrightarrow & \text{Gr}(1, \mathbb{C}^n) & \times \dots \times & \text{Gr}(i, \mathbb{C}^n) & \times \dots \times & \text{Gr}(n-1, \mathbb{C}^n) \\
 & & \downarrow & & \downarrow & & \downarrow \\
 & & \mathbb{P}^{n-1}(\mathbb{C}) & \times \dots \times & \mathbb{P}^{(n)}(i)(\mathbb{C}) & \times \dots \times & \mathbb{P}^{n-1}(\mathbb{C}).
 \end{array}$$

The action of GL_n

Invertible matrices

Sending one basis to any other means sending a flag to any other. Invertible matrices can send any flag to any other flag. The GL_n -orbit of a flag is the set of all flags.

Stabilisers of flags

Associating the basis $(v_1, \dots, v_n)$ to a flag, the stabilisers of the flag are the upper triangular matrices in that basis.

Flags in relative position

Position of two flags $U = (U_1 \subset U_2 \subset \dots)$ and $V = (V_1 \subset V_2 \subset \dots)$ in relation to each other

GL_n action

$GL_n = G$ can send the first flag anywhere, preserving some relationship between the flags (intersections, etc.)

Fix a standard basis $(v_1, v_2, \dots, v_n)$, and associated standard flag. The flag is the flag generated by the columns of upper triangular matrices.

Stabilizers of flags

Then the G -orbits of (V, U) are the same as the B -orbits of (F, U') , or just U' .

Flags in relative position

Stabilizers of flags

Then the G -orbits of (V, U) are the same as the B -orbits of U' .
The same as saying the G -orbits of

Explicit bijection

$$\begin{aligned} G(g_0B, g_1B) &\mapsto Bg_0^{-1}g_1B \\ G(B, gB) &\leftarrow BgB \end{aligned}$$

Indexing by the symmetric group

Every GL_n matrix can be written as BwB with $w \in S_n$,



Schubert cells

Cells indexed by S_n

The Schubert cells $\theta_w = BwB$ give a disjoint union of GL_n , S_n indexes the B-orbits.

Closure of Schubert cells

The closure $\overline{\theta_w}$ is closed by B-action, it is the union of Schubert cells.

Partial order

We say $u \leq w$ if $\theta_u \subset \overline{\theta_w}$.

Bruhat order

What does $u \leq w$ mean for the permutations u and w ?

Bruhat order for permutations

If $u(i+1) = w$ with $u(i) < u(i+1)$, then $u \leq w$

These conditions on u and w can be proved to be equivalent using arguments about the intersections with the standard flag, and taking limits to prove one cell is in the closure of another.

Structure of Schubert varieties

The Schubert varieties are the disjoint unions $\overline{\theta_w} = \sqcup_{u \leq w} \theta_u$.

Case of $\text{Gr}(2, 4)$

$w \in S_3$	$\theta_w \in G \setminus (G/B \times G/B)$		
	Lines	Planes	Intersection
$()$	$V_1 = U_1,$	$V_2 = U_2$	
(12)	$V_1 \neq U_1,$	$V_2 = U_2$	
(23)	$V_1 = U_1,$	$V_2 \neq U_2,$	$V_2 \cap U_2 = V_1 = U_1$
(123)	$V_1 \neq U_1,$	$V_2 \neq U_2,$	$V_2 \cap U_2 = V_1$
(132)	$V_1 \neq U_1,$	$V_2 \neq U_2,$	$V_2 \cap U_2 = U_1$
(13)	$V_1 \neq U_1,$	$V_2 \neq U_2,$	$V_2 \cap U_2 \notin \{V_1, U_1\}$

Table: Configurations of $(V_1 \subset V_2)$ and $(U_1 \subset U_2)$ in 3-dimensional projective space with associated permutations in S_3 .

Schubert cells as open balls

Dimension of a Schubert cell

The cell $\theta_w = BwB$ admits a unique expression as the B orbit of a row reduced matrix with a certain number of free entries a .

$$\theta_w \cong \mathbb{A}^a \cong B(\mathbb{C}^a, 1) \cong B(\mathbb{R}^{2a}, 1).$$

Proof sketch

The identity permutation corresponds to the identity matrix and has 0 free entries. If $w = u(i \ i + 1)$ and $u(i) < u(i + 1)$ then $\dim(\theta_w) = \dim(\theta_u) + 1$.

This means $u < w \implies \dim(\theta_w) < \dim(\theta_u)$ for the Bruhat order.

CW structure of the flag variety

CW-complexes

CW-complexes are made out of open balls of various dimensions which are glued onto lower-dimensional balls with continuous maps.

Gluing maps of Schubert cells

In our case the gluing maps are just the inclusion map $\partial\theta_w = \sqcup_{u < w} \theta_u \rightarrow \sqcup_{u < w} \theta_u$, since these $\dim(\theta_u) < \dim(\theta_w)$.

Cellular homology

Theory

Cellular homology provides an easy description of the homology groups of CW-complexes, especially when all cells are even-dimensional, in this case the group is a free group with the cells as a basis.

Homology of the flag variety

Given by classes $[\theta_w]$ of θ_w , depending only on the permutation w and not on the chosen basis.

Cup product

Definition

The cup product defines a ring operation on the cohomology group $\smile : H^i \times H^j \rightarrow H^{i+j}$.

Poincaré Duality

Let $[X], [Y]$ be Schubert classes. There is a open dense subset $\Omega \subset G$ such that $\forall g \in \Omega, [X] \smile [Y] = [X \cap gY]$.

Allows computation of intersections in general position.

The Grassmannian

We are interested in the Grassmannian $\text{Gr}(k, n)$.

Projection

We can send the flag to its k th subspace.

$$V_1 \subset V_2 \subset \dots \subset V_k \subset \dots \subset \mathbb{C}^n \mapsto V_k.$$

Schubert cells of the Grassmannian

The Schubert cells here are B -orbits of $\langle v_{w(1)}, v_{w(2)}, \dots, v_{w(k)} \rangle$.

Young diagrams

We are interested in the Grassmannian $\text{Gr}(k, n)$.

Definition

λ is a sequence of non-negative integers $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_i$.

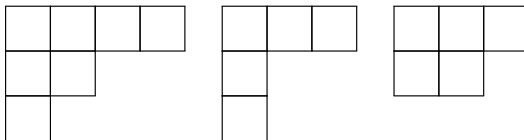


Figure: Young diagrams $(4 \geq 2 \geq 1)$, $(3 \geq 1 \geq 1)$, and $(3 \geq 2)$.

Correspondence with Young diagrams

Bijection

Associate $B\langle v_{w(1)}, v_{w(2)}, \dots, v_{w(k)} \rangle$ with the sequence a_i of $w(i)$ in increasing order.

$$1 \leq a_1 < \dots < a_k \leq n \rightarrow n - k \geq \lambda_1 \geq \dots \geq \lambda_k \geq 0$$

$$a_i \mapsto n - k + i - a_i$$

$$n - k + i - \lambda_i \leftarrow \lambda_i.$$

The number of boxes in the associated Young diagram corresponds to the homology group of $[\theta_w]$.

Young diagram correspondence of $\text{Gr}(2, 4)$

Young diagram	Cell	Notation	Geometric Meaning
	$B\langle v_1, v_2 \rangle$	$[X_0]$	Class of a line
	$B\langle v_1, v_3 \rangle$	$[X_1]$	Class of lines passing through a given point and contained in a given plane.
	$B\langle v_1, v_4 \rangle$	$[X_2']$	Class of lines passing through a point
	$B\langle v_2, v_3 \rangle$	$[X_2]$	Class of lines in a given plane
	$B\langle v_2, v_4 \rangle$	$[X_3]$	Class of lines intersecting some fixed line
$\emptyset$	$B\langle v_3, v_4 \rangle$	$[X_4]$	Class of all lines in space

Table: Schubert classes of $\text{Gr}(2, 4)$.

Pieri rule

Statement

Let (m) one line diagram with m boxes and λ be another Young diagram. Then $[X_{(m)}][X_\lambda] = \sum_{\mu \in m \otimes \lambda} X_\mu$, where $m \otimes \lambda$ is the set of all Young diagrams that fit into a $k(n-k)$ box that can be made by adding m boxes to λ , with no two boxes in the same column.

Examples

The Young diagrams in $1 \otimes (1)$ in a 2×2 box are  and .

The Young diagrams in $2 \otimes (2 \geq 1)$ in a 3×3 box are , , and .

Multiplication for cohomology ring of $\text{Gr}(2, 4)$

Product	Justification
$[X_4]$ is the multiplicative identity	Pieri Rule
$[X_3]^2 = [X_2] + [X_2']$	Pieri Rule
$[X_3][X_2] = [X_1]$	Pieri Rule
$[X_3][X_2'] = [X_1]$	Pieri Rule
$[X_3][X_1] = [X_0]$	Pieri Rule
$[X_3][X_0] = 0$	Pieri Rule
$[X_2]^2 = [X_0]$	Poincaré Duality
$[X_2][X_2'] = 0$	Poincaré Duality
$[X_2][X_1] = [X_3][X_2][X_2'] = 0$	$[X_1] = [X_3][X_2']$
$[X_2']^2 = [X_0]$	Poincaré Duality
$[X_2'][X_1] = [X_3][X_2][X_2'] = 0$	$[X_1] = [X_3][X_2]$
$[X_1]^2 = [X_3]^2[X_2]^2 = [X_3][X_3][X_0] = 0$	$[X_1] = [X_3][X_2]$
$[X_0][X_b] = 0 \quad \forall b \neq 4$	$[X_3][X_0] = 0$

Table: Multiplication table of the Schubert classes of $\text{Gr}(2, 4)$.

The computation

The computation

$$[X_4]^4 = [X_4]^2([X_2] + [X_2']) = [X_4]([X_1] + [X_1]) = 2[X_0]$$

$$[X_4]^4 = ([X_2] + [X_2'])^2 = [X_2]^2 + [X_2']^2 + 2[X_2][X_2'] = 2[X_0]$$

Gives that in general position, 4 lines in projective space have exactly 2 lines that intersect all four of them.